```
!pip install pandas
```

```
ERROR: Operation cancelled by user
```

```
import pandas as pd
```

```
from google.colab import files
uploaded = files.upload()
```

Choose Files No file chosen Upload widget is only available when the cell has been executed in the current browser session. Please rerun this cell to enable.

```
Saving Global_Superstore2.csv.zip to Global_Superstore2.csv.zip
```

```
import os
print(os.listdir())
```

```
['.config', 'Global_Superstore2.csv.zip', 'sample_data']
```

```
df = pd.read_csv(".config', 'Global_Superstore2.csv.zip', 'sample_data")
df.head()
```

```
---------------------------------------------------------------------------
FileNotFoundError                         Traceback (most recent call last)
/tmp/ipython-input-754/228408946.py in <cell line: 0>()
----> 1 df = pd.read_csv(".config', 'Global_Superstore2.csv.zip', 'sample_data")
      2 df.head()

                              4 frames
/usr/local/lib/python3.12/dist-packages/pandas/io/common.py in get_handle(path_or_buf, mode, encoding, compression, memory_map, is_text, errors, storage_options)
    871         if ioargs.encoding and "b" not in ioargs.mode:
    872             # Encoding
--> 873             handle = open(
    874                 handle,
    875                 ioargs.mode,

FileNotFoundError: [Errno 2] No such file or directory: ".config', 'Global_Superstore2.csv.zip', 'sample_data"
```

```
df = pd.read_csv("Global_Superstore2.csv.zip")
df.head()
```

```
---------------------------------------------------------------------------
UnicodeDecodeError                        Traceback (most recent call last)
/tmp/ipython-input-754/3698184434.py in <cell line: 0>()
----> 1 df = pd.read_csv("Global_Superstore2.csv.zip")
      2 df.head()

                              5 frames
parsers.pyx in pandas._libs.parsers.TextReader.__cinit__()

parsers.pyx in pandas._libs.parsers.TextReader._get_header()

parsers.pyx in pandas._libs.parsers.TextReader._tokenize_rows()

parsers.pyx in pandas._libs.parsers.TextReader._check_tokenize_status()

parsers.pyx in pandas._libs.parsers.raise_parser_error()

/usr/lib/python3.12/codecs.py in decode(self, input, final)

UnicodeDecodeError: 'utf-8' codec can't decode byte 0xe1 in position 6145: invalid continuation byte
```

```
import os
print(os.listdir())
```

```
['.config', 'Global_Superstore2.csv.zip', 'sample_data']
```

```
df = pd.read_csv("Global_Superstore.csv")
df.head()
```

```
---------------------------------------------------------------------------
FileNotFoundError                         Traceback (most recent call last)
/tmp/ipython-input-754/1411011137.py in <cell line: 0>()
----> 1 df = pd.read_csv("Global_Superstore.csv")
      2 df.head()

                              4 frames
/usr/local/lib/python3.12/dist-packages/pandas/io/common.py in get_handle(path_or_buf, mode, encoding, compression, memory_map, is_text, errors, storage_options)
    871         if ioargs.encoding and "b" not in ioargs.mode:
    872             # Encoding
--> 873             handle = open(
    874                 handle,
    875                 ioargs.mode,

FileNotFoundError: [Errno 2] No such file or directory: 'Global_Superstore.csv'
```

```
df = pd.read_csv("['.config', 'Global_Superstore2.csv.zip', 'sample_data']
")
df.head()
```

```
  File "/tmp/ipython-input-754/227744846.py", line 1
    df = pd.read_csv("['.config', 'Global_Superstore2.csv.zip', 'sample_data']
                     ^
SyntaxError: unterminated string literal (detected at line 1)
```

```
import os
print(os.listdir())
```

```
['.config', 'Global_Superstore2.csv.zip', 'sample_data']
```

```
df = pd.read_csv("'.config', 'Global_Superstore2.csv.zip', 'sample_data'")
df.head()
```

```
---------------------------------------------------------------------------
FileNotFoundError                         Traceback (most recent call last)
/tmp/ipython-input-754/1442676847.py in <cell line: 0>()
----> 1 df = pd.read_csv("'.config', 'Global_Superstore2.csv.zip', 'sample_data'")
      2 df.head()

                              4 frames
/usr/local/lib/python3.12/dist-packages/pandas/io/common.py in get_handle(path_or_buf, mode, encoding, compression, memory_map, is_text, errors, storage_options)
    871         if ioargs.encoding and "b" not in ioargs.mode:
    872             # Encoding
--> 873             handle = open(
    874                 handle,
    875                 ioargs.mode,

FileNotFoundError: [Errno 2] No such file or directory: "'.config', 'Global_Superstore2.csv.zip', 'sample_data'"
```

```
import pandas as pd
```

```
df = pd.read_csv("Global_Superstore2.csv.zip", compression='zip')
df.head()
```

```
---------------------------------------------------------------------------
UnicodeDecodeError                        Traceback (most recent call last)
/tmp/ipython-input-754/2589500042.py in <cell line: 0>()
      1 import pandas as pd
      2
----> 3 df = pd.read_csv("Global_Superstore2.csv.zip", compression='zip')
      4 df.head()

                              5 frames
parsers.pyx in pandas._libs.parsers.TextReader.__cinit__()

parsers.pyx in pandas._libs.parsers.TextReader._get_header()

parsers.pyx in pandas._libs.parsers.TextReader._tokenize_rows()

parsers.pyx in pandas._libs.parsers.TextReader._check_tokenize_status()

parsers.pyx in pandas._libs.parsers.raise_parser_error()

/usr/lib/python3.12/codecs.py in decode(self, input, final)

UnicodeDecodeError: 'utf-8' codec can't decode byte 0xe1 in position 6145: invalid continuation byte
```

```
import zipfile

with zipfile.ZipFile("Global_Superstore2.csv.zip", 'r') as zip_ref:
    print(zip_ref.namelist())
```

```
['Global_Superstore2.csv']
```

```
import pandas as pd
import zipfile

with zipfile.ZipFile("Global_Superstore2.csv.zip") as z:
    file_name = z.namelist()[0]   # get first file inside zip
    df = pd.read_csv(z.open(file_name), encoding='latin1')

df.head()
```

	Row ID	Order ID	Order Date	Ship Date	Ship Mode	Customer ID	Customer Name	Segment	City	State	...	Product ID	Category	Sub-Category	Product Name	Sales	Quantity	Discour
0	32298	CA-2012-124891	31-07-2012	31-07-2012	Same Day	RH-19495	Rick Hansen	Consumer	New York City	New York	...	TEC-AC-10003033	Technology	Accessories	Plantronics CS510 - Over-the-Head monaural Wir...	2309.650	7	0
1	26341	IN-2013-77878	05-02-2013	07-02-2013	Second Class	JR-16210	Justin Ritter	Corporate	Wollongong	New South Wales	...	FUR-CH-10003950	Furniture	Chairs	Novimex Executive Leather Armchair, Black	3709.395	9	0
2	25330	IN-2013-71249	17-10-2013	18-10-2013	First Class	CR-12730	Craig Reiter	Consumer	Brisbane	Queensland	...	TEC-PH-10004664	Technology	Phones	Nokia Smart Phone, with Caller ID	5175.171	9	0
3	13524	ES-2013-1579342	28-01-2013	30-01-2013	First Class	KM-16375	Katherine Murray	Home Office	Berlin	Berlin	...	TEC-PH-10004583	Technology	Phones	Motorola Smart Phone, Cordless	2892.510	5	0
4	47221	SG-2013-4320	05-11-2013	06-11-2013	Same Day	RH-9495	Rick Hansen	Consumer	Dakar	Dakar	...	TEC-SHA-10000501	Technology	Copiers	Sharp Wireless Fax, High-Speed	2832.960	8	0

5 rows × 24 columns

```
df
```

```python
df.head()
```

	Row ID	Order ID	Order Date	Ship Date	Ship Mode	Customer ID	Customer Name	Segment	City	State	...	Product ID	Category	Sub-Category	Product Name	Sales	Quantity	Discount
0	32298	CA-2012-124891	31-07-2012	31-07-2012	Same Day	RH-19495	Rick Hansen	Consumer	New York City	New York	...	TEC-AC-10003033	Technology	Accessories	Plantronics CS510 - Over-the-Head monaural Wir...	2309.650	7	0
1	26341	IN-2013-77878	05-02-2013	07-02-2013	Second Class	CR-12730	Craig Reiter	Corporate	Wollongong	New South Wales	...	FUR-CH-10003950	Furniture	Chairs	Novimex Executive Leather Armchair, Black	3709.395	9	0
2	25330	IN-2013-71249	17-10-2013	18-10-2013	First Class	CR-12730	Craig Reiter	Consumer	Brisbane	Queensland	...	TEC-PH-10004664	Technology	Phones	Nokia Smart Phone, with Caller ID	5175.171	9	0
3	13524	ES-2013-1579342	28-01-2013	30-01-2013	First Class	KM-16375	Katherine Murray	Home Office	Berlin	Berlin	...	TEC-PH-10004583	Technology	Phones	Motorola Smart Wireless Fax, High-Speed Phone, Cordless	2892.510	5	0
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
4	47221	SG-2013-4320	05-11-2013	06-11-2013	Same Day	RH-9495	Rick Hansen	Consumer	Dakar	Dakar	...	TEC-SHA-10000501	Technology	Copiers	Sharp Wireless Fax, High-Speed	2832.960	8	0

5 rows × 24 columns

51286	35398	US-2014-155768	20-06-2013	24-06-2013	Standard Class	ZC-21910	Zuschuss Carroll	Consumer	Houston	Texas	...	OFF-AP-10002906	Office Supplies	Appliances	Hoover Replacement Belt for Replacement Belt	0.444	1	0.7
51287	40470	US-2013-155768	02-12-2013	02-12-2013	Same Day	LB-16795	Laurel Beltran	Home Office	Oxnard	California	...	OFF-EN-10001219	Office Supplies	Envelopes	#10- 4 1/8" x 9 1/2" Security-Tint Envelopes	22.920	3	0
51288	9596	MX-2012-140767	18-02-2012	22-02-2012	Standard Class	RB-19795	Ross Baird	Home Office	Valinhos	São Paulo	...	OFF-BI-10000806	Office Supplies	Binders	Acco Index Tab, Economy	13.440	2	0.2
51289	6147	MX-2012-134460	22-05-2012	26-05-2012	Second Class	MC-18100	Mick Crebagga	Consumer	Tipitapa	Managua	...	OFF-PA-10004155	Office Supplies	Paper	Eaton Computer Printout Paper, 8.5 x 11	61.380	3	0

```python
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 51290 entries, 0 to 51289
Data columns (total 24 columns):
 #   Column          Non-Null Count  Dtype
---  ------          --------------  -----
 0   Row ID          51290 non-null  int64
 1   Order ID        51290 non-null  object
 2   Order Date      51290 non-null  object
 3   Ship Date       51290 non-null  object
 4   Ship Mode       51290 non-null  object
 5   Customer ID     51290 non-null  object
 6   Customer Name   51290 non-null  object
 7   Segment         51290 non-null  object
```

51290 rows × 24 columns

```
 8   City            51290 non-null  object
 9   State           51290 non-null  object
 10  Country         51290 non-null  object
 11  Postal Code     9994 non-null   float64
 12  Market          51290 non-null  object
 13  Region          51290 non-null  object
 14  Product ID      51290 non-null  object
 15  Category        51290 non-null  object
 16  Sub-Category    51290 non-null  object
 17  Product Name    51290 non-null  object
 18  Sales           51290 non-null  float64
 19  Quantity        51290 non-null  int64
 20  Discount        51290 non-null  float64
 21  Profit          51290 non-null  float64
 22  Shipping Cost   51290 non-null  float64
 23  Order Priority  51290 non-null  object
dtypes: float64(5), int64(2), object(17)
memory usage: 9.4+ MB
```

```
df.isnull().sum()
```

	0
Row ID	0
Order ID	0
Order Date	0
Ship Date	0
Ship Mode	0
Customer ID	0
Customer Name	0
Segment	0
City	0
State	0
Country	0
Postal Code	41296
Market	0
Region	0
Product ID	0
Category	0
Sub-Category	0
Product Name	0
Sales	0
Quantity	0
Discount	0
Profit	0
Shipping Cost	0
Order Priority	0

dtype: int64

```
numeric_cols = df.select_dtypes(include=['int64', 'float64']).columns

for col in numeric_cols:
    df[col] = df[col].fillna(df[col].mean())
```

```
categorical_cols = df.select_dtypes(include=['object']).columns

for col in categorical_cols:
    df[col] = df[col].fillna(df[col].mode()[0])
```

```
df.duplicated().sum()
```

```
np.int64(0)
```

```
df = df.drop_duplicates()
```

```
df['Order Date'] = pd.to_datetime(df['Order Date'], errors='coerce')
df['Ship Date'] = pd.to_datetime(df['Ship Date'], errors='coerce')
```

```
/tmp/ipython-input-754/3860214703.py:1: UserWarning: Parsing dates in %d-%m-%Y format when dayfirst=False (the default) was specified. Pass `dayfirst=True` or specify
  df['Order Date'] = pd.to_datetime(df['Order Date'], errors='coerce')
/tmp/ipython-input-754/3860214703.py:2: UserWarning: Parsing dates in %d-%m-%Y format when dayfirst=False (the default) was specified. Pass `dayfirst=True` or specify
  df['Ship Date'] = pd.to_datetime(df['Ship Date'], errors='coerce')
```

```
df.dtypes
```

	0
Row ID	int64
Order ID	object
Order Date	datetime64[ns]
Ship Date	datetime64[ns]
Ship Mode	object
Customer ID	object
Customer Name	object
Segment	object
City	object
State	object
Country	object
Postal Code	float64
Market	object
Region	object
Product ID	object
Category	object
Sub-Category	object
Product Name	object
Sales	float64
Quantity	int64
Discount	float64
Profit	float64
Shipping Cost	float64
Order Priority	object

dtype: object

```
df.columns = df.columns.str.replace(" ", "_")
```

```
df.info()
df.describe()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 51290 entries, 0 to 51289
Data columns (total 24 columns):
 #   Column          Non-Null Count  Dtype
---  ------          --------------  -----
 0   Row_ID          51290 non-null  int64
 1   Order_ID        51290 non-null  object
 2   Order_Date      51290 non-null  datetime64[ns]
 3   Ship_Date       51290 non-null  datetime64[ns]
 4   Ship_Mode       51290 non-null  object
```